

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458158.709 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.152 +/- 0.024
Transit depth (Rp/Rs)^2	2.3 +/- 0.72 [%]
Orbital Inclination (inc)	85.25 +/- 0.87
Airmass coefficient 1 (a1)	1.726 +/- 0.033
Airmass coefficient 2 (a2)	-0.39 +/- 0.13
Scatter in the residuals of the lightcurve fit is	1.88 %
Best Comparison Star	None
Optimal Aperture	2.29
Optimal Annulus	7.24
Transit Duration (day)	0.113 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.152 ± 0.024 vs 0.11 (deviation 1.52σ)
Duration fit vs published	0.113 d vs 0.1208 d (deviation 0.38σ)
Mid-time O–C	-2.6 min
Depth SNR	5.5
Residual scatter	1.88 %

Light curve

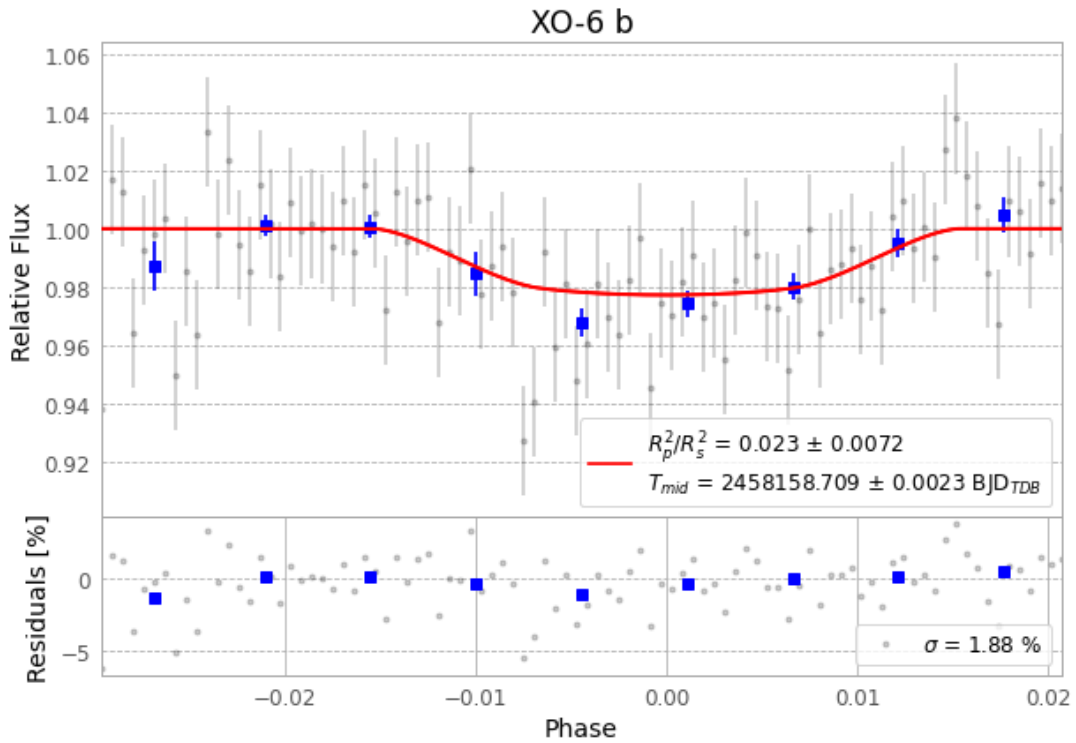


Figure 1 — FinalLightCurve_XO-6 b_2018-02-08.png (SHA-256 254d964db05270ee...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [515, 361], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e636504899856c52...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

92 FITS frames (60 MB), XO-6180209021512.FITS → XO-6180209064812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-XO-6-180209.log (3af4b4672c24...), FinalLightCurve_XO-6 b_2018-02-08.png (254d964db052...), FinalLightCurve_XO-6 b_2018-02-08.csv (dbd55073704f...)

Compute resources

Wall time 338.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-17 23:55Z.